

MD. MIFTAHUR RAHMAN SWAPNIL

Product Designer & Builder | AI, Frontend & Automation

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ABOUT ME

Product Designer & Builder with an innovative and curious mindset, always exploring new ideas, technologies, and ways to turn them into working products. I enjoy building and experimenting with new things, whether it is to solve a real-world problem, learn something new, or gain hands-on experience. I use AI as a practical part of my workflow to simplify tasks, improve user experiences, and reduce time spent on repetitive work. Currently, I'm especially interested in AI automation and building smarter workflows that make everyday processes easier and more efficient.

EDUCATION

Degree	Institution	Result	Year
B.Sc. in CSE	Bangladesh University of Business and Technology (BUBT)	CGPA 3.60/4.00	2023–2026 (Expected)
HSC — Science	Shaheed Police Smrity College, Mirpur-14, Dhaka	GPA 5.00/5.00	2021
SSC — Science	Kadirabad BL High School, Pirganj, Rangpur	GPA 5.00/5.00	2019

TECHNICAL SKILLS

Frontend: Next.js, React, TypeScript, JavaScript

Backend & Database: Supabase, PostgreSQL, REST APIs

AI & Automation: AI APIs, n8n, AI-assisted development, workflow automation

Other: Git, GitHub, etc.

FEATURED PROJECTS

Edu51Portal

Next.js 14, TypeScript, Supabase PostgreSQL, Google Drive API, Tailwind CSS

- Designed and launched an all-in-one academic platform serving ~100 active BUBT students, delivering instant routine lookups, exam notices, and course materials.
- Integrated Google Drive API with Supabase to distribute academic PDFs securely with \$0 cloud storage infrastructure overhead.
- Engineered section-specific schedule filtering, fuzzy course search, and mobile-first responsive interfaces with sub second page performance.

PawfectMatch

React Native, Node.js, Supabase Realtime DB, AI Pet Care Assistant(Gemini Api)

- Designed and developed a pet adoption mobile platform connecting verified rescuers with adopters, featuring AI-guided pet care recommendations and matchmaking.

Personal AI Assistant – Mikasa

Node.js, Telegram Bot API, Supabase, n8n, Multi-LLM Orchestration, Local PC Bridge

- Architected an autonomous personal AI operating system with persistent memory extraction, continuous learning, and multi-model failover (Gemini 2.5 Flash & OpenRouter).
- Telegram bot controls, cross-platform cron automation, and secure local PC execution bridges for terminal and desktop operations.
- Integrated real-time database grounding with Supabase.

MuteBD

Flutter, Dart, Google Maps API, Background Geofencing, Firebase

- Engineered a location-based mobile application that automatically silences smartphones in designated quiet zones (mosques, hospitals, libraries, classrooms) using GPS geofencing.
- Implemented battery-optimized background geofence listeners with customizable boundary radii and offline-first quiet zone caching.

OpusGen AI

Next.js, TypeScript, fal.ai (Flux/Gemini), Supabase, Vercel

- OpusGenAI is an AI-powered product photography studio that helps businesses turn ordinary product photos into professional marketing visuals without requiring a physical photography studio, professional camera equipment, or advanced design skills.
- AI-generated image, video and captions for marketing and social media use.

RESEARCH & PUBLICATIONS**Relation-Aware Graph Retrieval over a Curriculum Knowledge Graph for Prerequisite QA (Accepted (IEEE OMLET 2026))**

- Authors: Md. Jahidul Kamal Islam, **Md. Miftahur Rahman Swapnil**, Md. Asif Ali, Shrabani Das, Shefayatuj Johara Chowdhury.
- Engineered a curriculum knowledge graph RAG system with a 2-layer R-GCN encoder and DistMult decoder over 418 nodes and 558 typed edges verified in Neo4j.
- Achieved 45.5% exact-set match (vs. 22.7% text-RAG), 63.8% Entity F1, and a 100% correct abstention rate on unanswerable questions with 0 query-time LLM overhead.

AI-Enabled Smart Classroom Monitoring and Safety Automation (Accepted (IEEE i-COSTE 2026))

- Authors: Md. Jahidul Kamal Islam, **Md. Miftahur Rahman Swapnil**, Md. Asif Ali, Nishat Anjum Sara, Sheikh Shamia Hasan Nila, Sadah Anjum Shanto, (Assistant Professor).
- Engineered an IoT smart classroom safety automation system integrating sensor telemetry (DHT11, PIR, Ultrasonic, Sound) with automated door actuators and alert buzzers.
- Implemented dual operational presets (Normal Mode & Lecture Mode) integrated with Firebase Realtime Database and mobile dashboard telemetry.

LEADERSHIP & COMMUNITY ENGAGEMENT**BASIS Students' Forum of BUBT Chapter Club**

General Member, Graphics Designer & Media and Publication Secretary

- Directed creative visual branding, digital promotional media, and publication outreach for tech seminars, workshops, and industry networking events.

LANGUAGES

- **Bangla** — Native / First Language
- **English** — Professional Working Proficiency